

Concerted nucleophilic aromatic substitutions

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Nucleophilic aromatic substitution (S_NAr) is one of the most widely applied reaction classes in pharmaceutical and chemical research, providing a broadly useful platform for the modification of aromatic ring scaffolds. The generally accepted mechanism for S_NAr reactions involves a two-step addition–elimination sequence via a discrete, non-aromatic Meisenheimer complex. Here we use $^{12}C/^{13}C$ kinetic isotope effect (KIE) studies and computational analyses to provide evidence that prototypical S_NAr reactions in fact proceed through concerted mechanisms. The KIE measurements were made possible by a new technique that leverages the high sensitivity of ^{19}F as an NMR nucleus to quantitate the degree of isotopic fractionation. This sensitive technique permits the measurement of KIEs on 10 mg of natural abundance material in one overnight acquisition. As a result, it provides a practical tool for performing detailed mechanistic analyses of reactions that form or break C–F bonds.

Whether a given chemical transformation proceeds through a stepwise or concerted mechanism is a question of both fundamental interest and practical importance, with direct implications for the stereospecificity, sensitivity to medium effects, and product distribution of the reaction¹. In general, stepwise pathways are favoured when productive reactive intermediates are energetically accessible. For example, the isolation and characterization of the direct products of nucleophilic addition to aromatic rings, termed Meisenheimer complexes (Fig. 1a), have led to the widely held notion that S_NAr reactions² generally proceed via stepwise addition–elimination mechanisms^{1,2}.

The participation of a discrete Meisenheimer complex as an intermediate in an S_NAr mechanism requires that the anionic adduct be more thermodynamically stable than the transition state (TS) for the concerted pathway, as well as possess sufficient kinetic stability to have a significant lifetime before elimination. Nearly all known Meisenheimer complexes contain both strongly electron-withdrawing substituents, such as nitro groups, and poor leaving groups, such as fluoride, that are expected to retard elimination³. In the S_NAr reactions that are typically employed in pharmaceutical synthesis⁴, which involve less stabilized anions or good leaving groups, the intermediates are not detectable but have nonetheless been assumed to exist².

However, recent theoretical and experimental work, most notably from the Ritter group^{5,6}, has challenged this assumption. More broadly, concerted nucleophilic substitution reactions at sp^2 carbons have long been established for reactions of vinyl halides^{7–11}, as well as acyl^{12–24}, phosphorus^{25–29} and sulfur group transfer reactions^{30–34}. In the context of S_NAr reactions, it has been shown that the quasi-symmetric¹ addition of substituted phenoxides to phenoxytriazines is concerted^{35,24}, while the apparently analogous addition of pyridines to pyridyltriazines is stepwise^{36,37}. For many other reactions, computations^{38–49} predict concerted mechanisms, contradicting the textbook view that S_NAr reactions proceed via stepwise addition–elimination mechanisms⁵⁰.

We sought to use experiments to ascertain whether stepwise mechanisms are indeed generally operative in S_NAr reactions by studying three transformations with putative intermediates that would be strongly, moderately or weakly stabilized (Fig. 1b). Structure **A** is strongly stabilized by nitro group substituents and poor leaving groups (fluoride and methoxide), and related

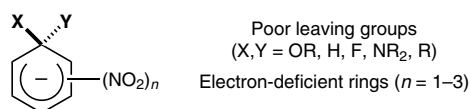
reactions have been demonstrated to be stepwise in nature⁵¹. In contrast, structure **B** is only weakly stabilized by its nitrogen-containing heterocycle and ester substituent and contains a good leaving group (bromide)—factors that might be expected to favour a concerted reaction. Reaction **C** represents a borderline case, in which structure **C** is stabilized by strongly electron-withdrawing substituents, but destabilized by a good leaving group (chloride).

Kinetic isotope effects (KIEs) can provide direct insight into the bonding in TSs, and are therefore useful tools for the evaluation of reaction concertedness¹. In S_NAr reactions, the primary $^{12}C/^{13}C$ KIE at the carbon undergoing substitution is expected to provide the most useful information. One powerful strategy for measuring such KIEs is to use quantitative ^{13}C NMR spectroscopy, ideally at natural isotopic abundance⁵². However, the high precision required for the determination of intrinsically small (<10%) heavy-atom KIEs, combined with the low natural abundance and gyromagnetic ratio of the ^{13}C nucleus, renders this type of experiment generally challenging to implement. When the carbons of interest are not bound to a proton, as is the case for the sites undergoing substitution in S_NAr reactions, dipolar relaxation is particularly inefficient. This requires long delays (minutes) between scans and further reduces sensitivity.

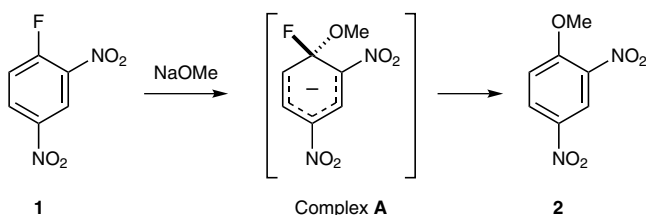
Results

We envisioned an alternative approach for the quantification of ^{13}C in the specific case of transformations involving C–F bond formation or cleavage, which are characteristic of many S_NAr reactions. In 1H -decoupled ^{19}F NMR spectra of organofluorine compounds at natural abundance, resonances appear as a large ^{12}C – ^{19}F parent singlet flanked by a small ^{13}C – ^{19}F satellite doublet (Fig. 2a). In principle, integration of these satellite integrals could provide a highly sensitive means of determining $^{12}C/^{13}C$ isotope fractionation because the signal-to-noise ratio (SNR) of each ^{13}C – ^{19}F satellite is theoretically 13.6 times greater (per fluorine) than that of the corresponding peak in a ^{13}C spectrum. This increase in sensitivity translates to a potential 185-fold reduction in acquisition time⁵³.

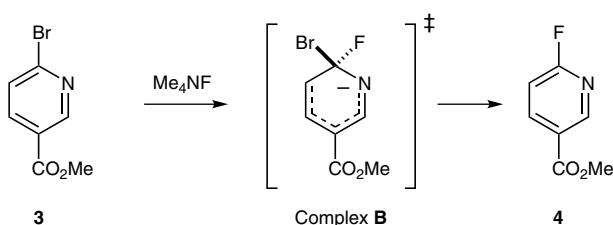
Unfortunately, it is not possible to obtain accurate satellite integrals directly⁵⁴ because of interference from the much larger parent peak, as well as possible overlapping impurities (Fig. 2b). However, we found that accurate satellite integrals could be obtained from

a Previously studied Meisenheimer complexes**b** This study

Reaction A (stepwise)



Reaction B (concerted)



Reaction C (borderline)

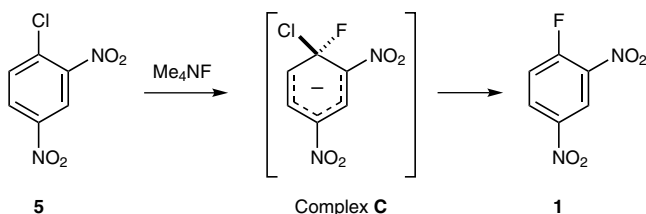


Fig. 1 | Scope of study. **a**, Previously observed Meisenheimer complexes have always been highly stabilized by poor leaving groups and electron-deficient rings. **b**, The S_NAr reactions studied here. Bold letters denote Meisenheimer complexes, which span a range of stabilities. (The term ‘complex’ does not necessarily indicate that the structure is a true intermediate.)

multiple-quantum-filtered (MQF) spectra in which the ^{12}C signals are suppressed (Fig. 2c)⁵³. This technique was validated by determining the $^{12}\text{C}/^{13}\text{C}$ KIE for a simple nucleophilic substitution reaction using both the traditional ^{13}C NMR and new MQF methods (Table 1). The MQF method was found to require far less material (as little as 10 mg with overnight acquisition; Table 1, entry 5) and acquisition time (under 1 h with 50 mg of material; Table 1, entry 6). The consensus value of 1.059(3) determined by the new method agrees with the predicted S_N2 KIE of 1.057 (M11/jun-cc-pVTZ/PCM). Consistent with earlier studies^{55,56}, the predicted KIE is insensitive to both TS geometry and computational method (Supplementary Section 4b).

The greatly increased sensitivity of the MQF method results from the larger gyromagnetic ratio and shorter T_1 relaxation times of ^{19}F over ^{13}C nuclei. Sensitivity can be further increased by using the parent $^{12}\text{C}-^{19}\text{F}$ peak from a routine $^{19}\text{F}\{^1\text{H}\}$ spectrum, rather than a different $^{13}\text{C}-^{19}\text{F}$ satellite in the MQF spectrum, as an internal reference⁵⁷ (Table 1, entries 3–6). Additionally, because a fluorination or defluorination reaction generally involves the appearance or disappearance of a well-separated ^{19}F resonance, unpurified material can be used (Table 1, entry 4).

Table 1 | Comparison of KIE measurements

	Method	Time (h)	KIE (s.e.)
1	Singleton (300 mg) ^a	9.9	1.058(6), 1.060(6)
2	MQF (50 mg) ^b	4.6	1.057(5), 1.065(6)
3	MQF (50 mg)	5.2	1.057(3), 1.062(4)
4	MQF (50 mg) ^c	5.2	1.059(4), 1.060(4)
5	MQF (10 mg)	9.9	1.061(6)
6	MQF (50 mg) ^d	0.7	1.055(10)
Consensus (s.d.):			1.059(3)

Samples are pure unless otherwise noted. Acquisition times are given for each pair of partial and full conversion samples. Pairs of KIEs refer to independent chemical replicates. Error bars (in parentheses) refer to standard errors (s.e.) of the mean (t distribution) and reflect technical variation due to errors in the measurement of conversion and satellite area. KIEs are referenced to $^{12}\text{C}-^{19}\text{F}$ unless otherwise noted. s.d., standard deviation. ^aKIEs are referenced to ^{13}C . ^bKIEs are referenced to $^{13}\text{C}-^{19}\text{F}$. ^cUnpurified sample. ^dShigemi tube used.

The accuracy and sensitivity of the MQF method allowed us to determine the $^{12}\text{C}/^{13}\text{C}$ KIEs for several S_NAr fluorination and defluorination reactions. To provide a theoretical basis for interpreting these measurements, we assessed the accuracy of various computational methods in describing the potential energy surfaces of reactions A, B and C. Using geometries spanning a range of carbon–nucleophile and carbon–leaving group distances, we found that B3LYP-D3(BJ)/jun-cc-pVTZ⁵⁸ most closely reproduced the energies calculated using the benchmark coupled cluster method DLPNO-CCSD(T)⁵⁹/aug-cc-pVTZ (Supplementary Sections 4c–e).

The surface for reaction A in implicit solvent (polarizable continuum model; PCM) clearly shows an intermediate (Fig. 3a), and is consistent with the second step being rate-determining. The experimental KIE of 1.035(3) agrees closely with the calculated KIE of 1.033. This provides strong evidence for the validity of the stepwise mechanism in the case of reaction A.

In contrast, computational analysis of reaction B reveals no evidence of an intermediate, either on its energy surface (Fig. 3c) or intrinsic reaction coordinate (Fig. 3e). Additionally, quasiclassical⁶⁰ dynamics initialized from TS B proceed quickly to product (mean time, 91 fs) and nearly all trajectories are productive (94%), indicating that the reaction is truly concerted. The predicted KIE of 1.039 is in reasonable agreement with the experimental value of 1.035(3).

Discussion

Although the experimentally determined KIEs for stepwise reaction A and concerted reaction B are the same (1.035), the calculations reveal that those values are in fact of very different magnitudes relative to the maximum values possible for the two reactions. For a given reaction, a ‘KIE surface’ (for example, Fig. 3b) can be generated by predicting KIEs for each point on the energy surface (for example, Fig. 3a). The largest value on this KIE surface may then be defined as the maximum KIE. When the bonds in the ground state are strong, more vibrational energy can be lost in the TS, resulting in a larger maximum KIE. In reaction A, a strong C–F bond is broken, and the maximum KIE is 1.070 (Fig. 3b, red). By contrast, in

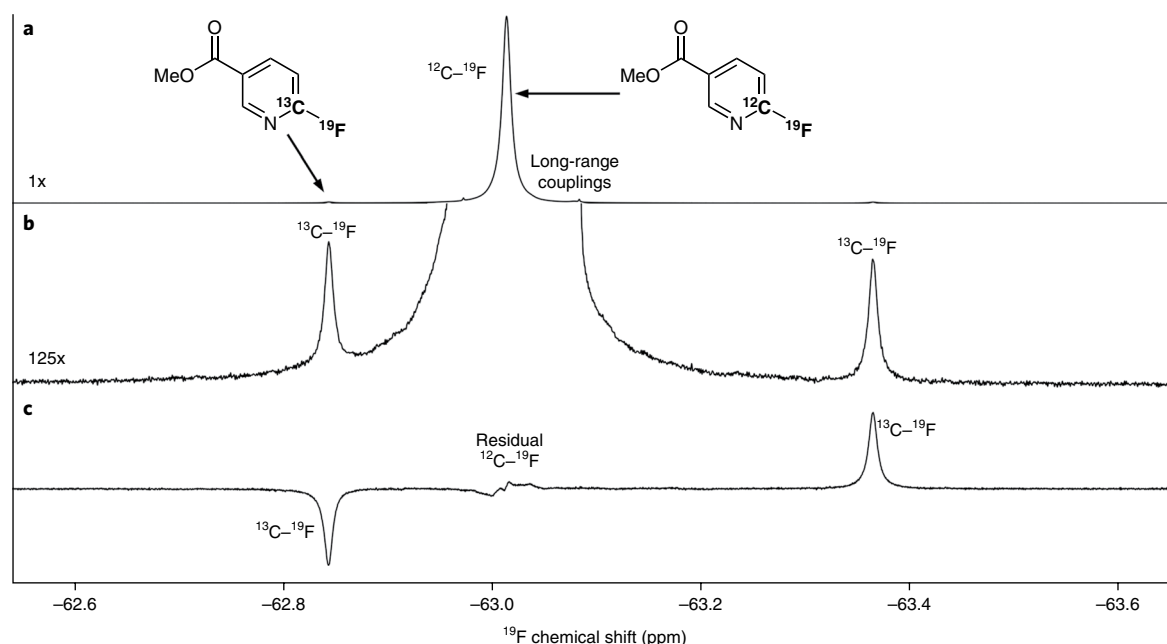


Fig. 2 | Assessing ^{13}C isotopic fractionation by suppressing NMR signals from fluorine atoms bound to ^{12}C . **a**, Standard $^{19}\text{F}\{^1\text{H}\}$ spectrum showing the parent $^{12}\text{C}-^{19}\text{F}$ peak flanked by two $^{13}\text{C}-^{19}\text{F}$ satellites. **b**, The standard spectrum in **a**, enlarged 125x. Accurate satellite integrals cannot be obtained directly due to overlap with the parent peak. **c**, MQF $^{19}\text{F}\{^1\text{H}\}$ spectrum. Suppression of the parent peak allows accurate integration of the satellites.

reaction B, a weak C–Br bond is broken, and the maximum KIE is 1.045 (Fig. 3d, green).

The maximum KIE in any particular reaction is obtained when the bonds to both the nucleophile and electrophile are weakest in the TS. Thus, the size of the actual KIE relative to its theoretical maximum is diagnostic of a stepwise or concerted mechanism. Stepwise mechanisms alter one bond at a time, leading to small KIEs (47% of the maximum value for reaction A). Conversely, concerted mechanisms alter both bonds simultaneously, passing through a nearly symmetric region in which both bonds are relatively weak, leading to large KIEs (87% of the maximum value for reaction B). This correspondence between small relative KIEs for stepwise mechanisms and large relative KIEs for concerted mechanisms parallels the behaviour of $\text{S}_{\text{N}}2$ reactions⁶¹.

Reaction C is a borderline case. Once again, the experimentally determined $^{12}\text{C}/^{13}\text{C}$ KIE of 1.045(3) is in good agreement with the predicted KIE of 1.040. This prediction is 73% of the maximum calculated value of 1.055, and lies between the values for the stepwise (47%) and concerted (87%) pathways. Borderline behaviour is observed because the putative intermediate would be stabilized by two electron-withdrawing groups (NO_2), but destabilized by a good leaving group (Cl). Accordingly, the minimum energy path does not pass through an intermediate, even though structures in the vicinity of the Meisenheimer complex are relatively stable (Fig. 3f). As a result, trajectories exiting the formal TS encounter this shallow region and linger for multiple vibrations before passing to product (mean time of 233 fs). The reaction mechanism can therefore be viewed either as concerted, with a long-lived TS, or stepwise, with a short-lived intermediate.

The transition from stepwise to concerted behaviour can be rationalized using qualitative Marcus theory^{62–64}. When the Meisenheimer complex is lower in energy than the intersection of the potential energy surfaces of the starting materials and products, the reaction is stepwise (Fig. 4a). When the intermediate is higher in energy than the intersection, the reaction mechanism is concerted with the TS resembling the minimum of the Meisenheimer curve (Fig. 4b). In borderline cases, such as when a stabilized anion

is adjacent to a good leaving group, the Meisenheimer complex may intersect the reaction coordinate as a shallow minimum or shoulder (Fig. 4c).

At first glance, concerted $\text{S}_{\text{N}}\text{Ar}$ mechanisms may appear difficult to rationalize on stereoelectronic grounds given that backside attack by the nucleophile on the C–X σ^* orbital of an aryl halide is precluded⁶⁵. However, this argument assumes that the bonding in the TS resembles that of the starting material. In fact, concerted transition structures actually possess characteristics of a Meisenheimer intermediate. For example, charge⁶⁶, nucleus-independent chemical shift (NICS)⁶⁷ and natural bond orbital (NBO)⁶⁸ analyses all confirm that TS B is essentially a delocalized, but non-aromatic, anion (Fig. 3e). This anion is generated by two concurrent, but asynchronous processes: C–F bond formation by donation of the fluoride lone pair into the C=N π^* orbital, and C–Br bond cleavage by donation of the incipient nitrogen lone pair into the C–Br σ^* orbital (Fig. 4d). The result can be viewed as a ‘Meisenheimer transition state’.

The qualitative picture of a continuum between stepwise and concerted $\text{S}_{\text{N}}\text{Ar}$ mechanisms depicted in Fig. 4 is supported quantitatively by the close agreement between the theoretical and experimental KIEs. To evaluate the overall prevalence of concerted $\text{S}_{\text{N}}\text{Ar}$ pathways, we carried out a computational survey of 120 $\text{S}_{\text{N}}\text{Ar}$ reactions spanning a variety of typical ring types, nucleophiles, and leaving groups using B3LYP/6-31+G*/PCM(DMSO). This more economical level of theory also corresponds closely to the coupled cluster energy surface (Supplementary Section 4g), and similar outcomes would be expected from other density functional theory (DFT) methods. Remarkably, 99 of these reactions (83%) are predicted to proceed via concerted mechanisms.

Where an $\text{S}_{\text{N}}\text{Ar}$ reaction lies on the stepwise/concerted continuum is determined primarily by the structural features of the reactants, rather than by the rate of their reaction. For nucleophilic substitutions on aryl rings, stepwise mechanisms are only predicted to occur when both a strongly electron-withdrawing substituent (for example, nitro) is present and fluoride is the nucleophile or leaving group. For substitutions on pyridine, pyrazine and pyrimidine, no stepwise mechanisms are predicted. Given that many $\text{S}_{\text{N}}\text{Ar}$ reactions

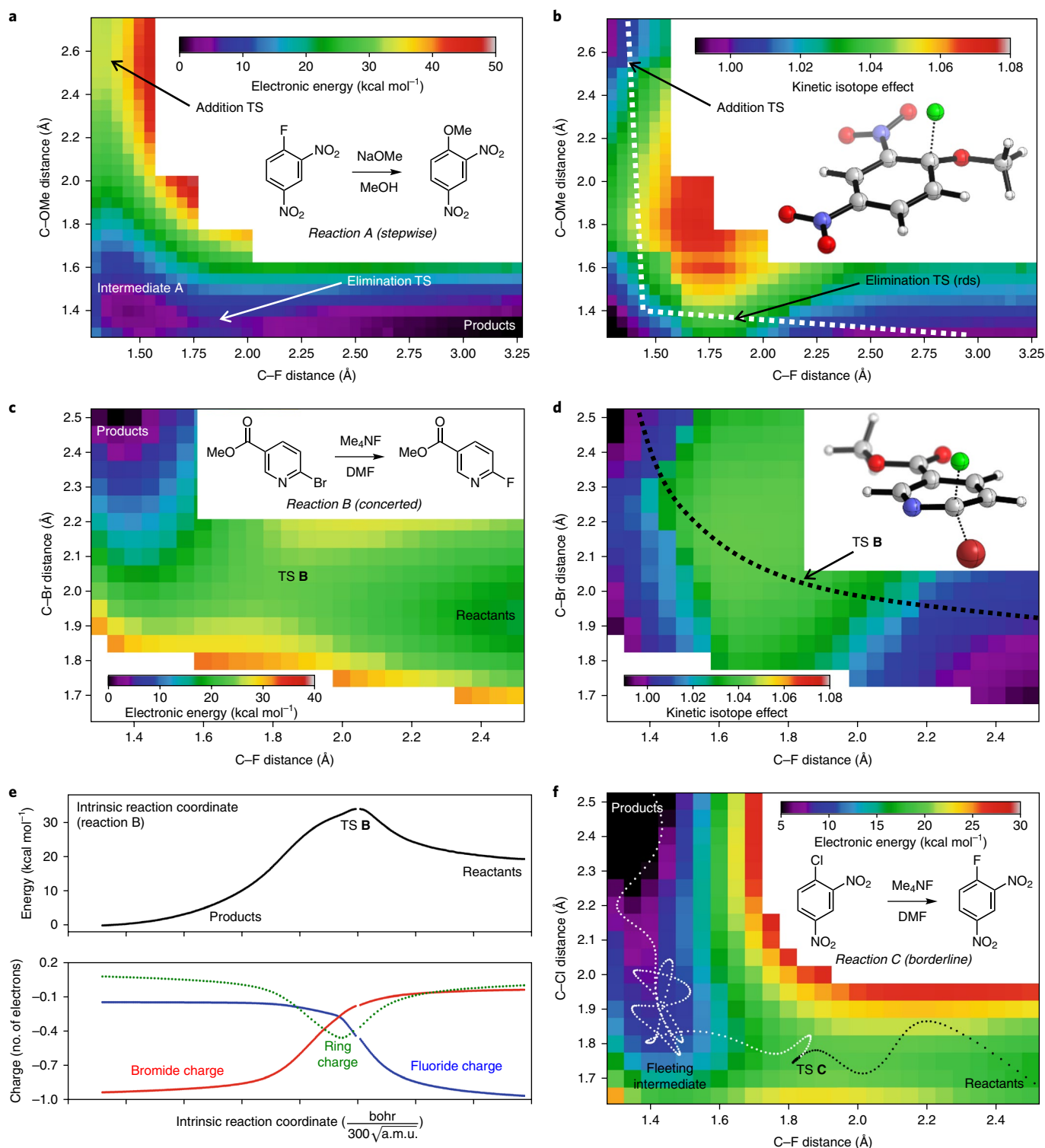


Fig. 3 | Computational analysis of the transition from stepwise to concerted behaviour (B3LYP-D3(BJ)/jun-cc-pVTZ). **a**, Potential energy surface for reaction A. An intermediate is apparent (lower left). **b**, Predicted KIE as a function of geometry for reaction A. The lowest-energy stepwise mechanism follows the white path, avoiding the region of large KIEs. rds, rate-determining step. **c**, Potential energy surface for reaction B. The 'Meisenheimer region' is very high in energy (white). **d**, Predicted KIE as a function of geometry for reaction B. The lowest-energy concerted mechanism follows the black path, resulting in a KIE that approaches the maximum possible value. **e**, Potential energy (top) and charge distribution (bottom) along the intrinsic reaction coordinate for reaction B. Negative charge is distributed between the nucleophile, the leaving group and the ring in the non-aromatic TS. Breaks in the curves indicate the position of TS B. **f**, Potential energy surface for reaction C. A typical trajectory (dotted path) reflects a fleeting intermediate or long-lived TS in the Meisenheimer region (lower left).

of interest are performed on heterocycles with good leaving groups (for example, Cl, Br)^{1,2}, it is likely that concerted mechanisms are actually very common.

The mechanistic analysis described herein was enabled by the sensitivity and practicality of the MQF method for determining ¹²C/¹³C KIEs. Our observations confirm that the nucleo-

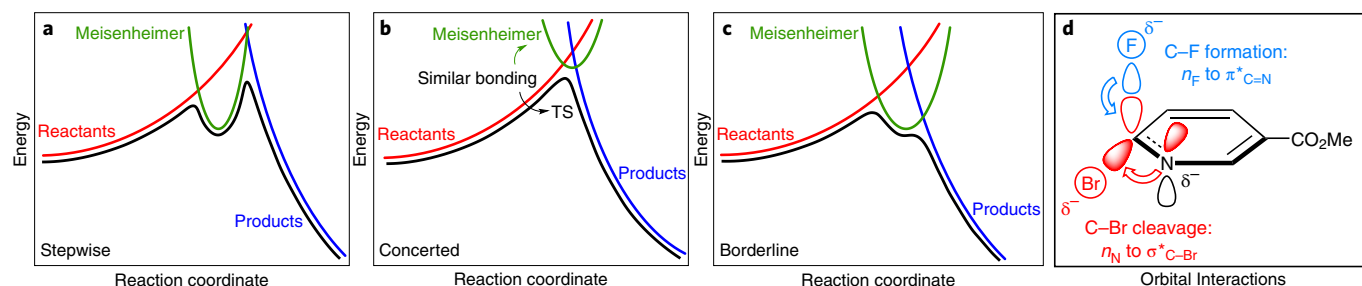


Fig. 4 | Simplified Marcus analysis of stepwise versus concerted S_NAr reactions. **a**, In stepwise reactions, the Meisenheimer structure is highly stabilized, leading to a minimum along the reaction coordinate (reaction A). **b**, Concerted reactions result if the Meisenheimer structure is less stable. This is the typical situation (reaction B). **c**, When the Meisenheimer structure is highly stabilized, but leaving group elimination is facile, a borderline situation (reaction C) results. **d**, Meisenheimer TSs are stabilized by concurrent donor–acceptor interactions (TS B is shown).

philic adducts identified in refs ⁶⁹ and ⁷⁰ over a century ago are indeed involved in S_NAr reactions, but often as TSs rather than intermediates. We anticipate that the MQF method will enable the study of other fluorination and defluorination pathways of synthetic interest.

Methods

Synthesis. Reactions were carried out in round-bottomed flasks under nitrogen. Commercially available reagents were purchased and used as received unless otherwise noted. The conversion of reactions was determined by quantitative ^{19}F NMR spectroscopy using *p*-difluorobenzene as an internal standard, with the exception of reaction A, which was analysed by HPLC using naphthalene as an internal standard. Products and recovered starting materials were purified by column chromatography, except for ‘impure’ samples, which were only subjected to an aqueous extraction. For detailed procedures and spectroscopic characterization data, see Supplementary Section 1.

Isotope effect methodology. NMR samples greater than 10 mg in mass were prepared in Wilmad 528-PP-9 tubes and flame-sealed under air at room temperature. The 10 mg samples were prepared in Shigemi tubes and sealed with parafilm. All spectroscopic measurements were performed at 25 °C on a Varian Inova 500 MHz machine fitted with an indirect detection probe (HFC) and a gradient driver. Specific details of the MQF pulse sequence are provided in Supplementary Section 3 and the PyKIE repository. For ^{12}C -referenced experiments, standard ^{19}F and MQF experiments were interwoven (for example, MQF then ^{19}F then MQF). This pattern was repeated throughout the duration of the measurement.

NMR data were processed using Python and the nmrglue package⁷¹ using the PyKIE pipeline. Each free induction decay (FID) was zero-filled (4 $\times$), apodized (exponential), Fourier transformed and baseline corrected. Peaks were picked manually and centred on regions of interest. Widths of the peak-containing regions of interest were held constant as follows: MQF, 0.045 ppm; ^{19}F , 0.15 ppm; 1H , 0.08 ppm; ^{13}C , 0.12 ppm and 0.14 ppm. SNRs were calculated for each peak by dividing the signal intensity at half height by the root-mean-square deviation of the signal in a noisy region.

Each KIE measurement was made by comparing the fractionation between a partial conversion sample and a full conversion sample (for product-based analyses) or between recovered and unreacted starting material. When multiple samples were available, data for the full conversion (or unreacted starting material) sample were pooled to give two independent estimates of the KIE. Values reported in the text correspond to the average of these estimates. Error bars for individual measurements correspond to standard errors and were propagated from known standard deviations in conversion and integral area. Detailed procedures and Excel spreadsheets for reproducing these calculations are provided in Supplementary Section 3 and the PyKIE repository.

Calculations. DFT calculations were performed with Gaussian 09 or Gaussian 16⁷². Bigeleisen–Mayer KIE predictions were made using PyQuiver⁷³. Tunnelling corrections were calculated using the one-dimensional Bell method. For the M06-2 $\times$ prediction of the S_N2 isotope effect, a multidimensional CVT/SCT correction was also calculated using GAUSSRATE⁷⁴/POLYRATE⁷⁵ and found to agree closely with the Bell correction. Coupled cluster calculations were carried out using ORCA 4.0.0 using TightPNO cutoffs⁷⁶.

Direct quasiclassical trajectory studies were carried out on the B3LYP-D3(BJ)/6-31+G*/PCM surface as previously described. Trajectories were initialized from a thermal distribution (298 K) of vibrational states using the position eigenstates of a quantum harmonic oscillator. No displacements were made in any

modes with a frequency of less than 50 cm⁻¹, including the transition vector. The sign of the velocity in all modes was randomized except for the transition vector, which was directed towards the product. Timesteps of 1.0 fs were calculated by the velocity Verlet method and each simulation was run for 500 fs in both the forward and reverse directions.

The predicted KIEs for reactions A, B and C were calculated with B3LYP-D3(BJ)/may-cc-pVQZ at 298 K (jun-cc-pVTZ is extremely similar) using PCM (that is, IEFPCM) implicit solvation and a scaling factor^{77–79} of 1.2 for the sphere radii. The addition of explicit methanol molecules to reaction A or explicit water molecules to reaction B had a negligible effect. The inclusion of at least one explicit water molecule was crucial for reaction C. The effect of solvation parameters is examined thoroughly in Supplementary Section 4f.

The energy and KIE surfaces shown in Fig. 3a–d,f were calculated at B3LYP-D3(BJ)/jun-cc-pVTZ/PCM with no explicit solvent molecules. The intrinsic reaction coordinate energetics and charges shown in Fig. 3e were calculated at B3LYP-D3(BJ)/6-31+G*/PCM. Each coloured square in Fig. 3 corresponds to a structure in which the carbon–nucleophile and carbon–electrophile bond distances have been held frozen, but all other geometric parameters have been allowed to relax. Bigeleisen–Mayer KIEs were then calculated for each grid point using the harmonic frequencies of the isolated substrate and the grid point geometry at 298 K. The maximum KIE was then taken as the maximum predicted KIE over the entire grid. The trajectory overlaid on Fig. 3f was calculated with an explicit water molecule at B3LYP-D3(BJ)/6-31+G*. All calculations in Fig. 3 used default PCM settings.

A total of 120 S_NAr reactions spanning a range of ring types (benzene, (NO₂)_{m=1,2,3}-benzene, H₃C–C=O–benzene, pyridine, pyrazine, pyrimidine), nucleophiles (F⁻, MeO⁻, N₃⁻, Me₂N⁻, formate⁻) and leaving groups (F⁻, Cl⁻, Br⁻) were thoroughly examined at B3LYP-D3(BJ)/6-31+G*/PCM(DMSO). Consistent with previous studies^{38–49}, in all but 21 cases it was not possible to locate a Meisenheimer intermediate by beginning optimizations in the expected neighbourhood of the intermediate. Intrinsic reaction coordinate (IRC) searches were attempted for each reaction. In no case did any converged IRC calculation show the unexpected existence of an intermediate. For the significant number of remaining unconverged cases, growing string⁸⁰ calculations were performed. Similarly, no string calculations revealed the unexpected existence of an intermediate. TSs and intermediates were located for nearly all of the reactions and each TS was confirmed to connect to the expected starting materials and products by the means above. In a small number of cases involving highly exothermic reactions it was not possible to locate saddle points on the potential energy surface (for details see Supplementary Section 4g).

Data availability. User-friendly software pipelines that can be used to measure and predict KIEs are freely available at www.github.com/ekwan/PyKIE and www.github.com/ekwan/PyQuiver. Raw NMR spectra and computed quasiclassical trajectories are available from the corresponding author upon reasonable request. All other data supporting the findings of this study are available within the Article and its Supplementary Information files.

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References

- Williams, A. *Concerted Organic and Bio-Organic Mechanisms* (CRC, Boca Raton, FL, 1999).
- Terrier, F. *Modern Nucleophilic Aromatic Substitution* (Wiley-VCH, Weinheim, 2013).
- Terrier, F. Rate and equilibrium studies in Jackson–Meisenheimer complexes. *Chem. Rev.* **82**, 77–152 (1982).

4. Brown, D. G. & Boström, J. Analysis of past and present synthetic methodologies on medicinal chemistry: where have all the new reactions gone? *J. Med. Chem.* **59**, 4443–4458 (2016).
5. Neumann, C. N., Hooker, J. M. & Ritter, T. Concerted nucleophilic aromatic substitution with $^{19}\text{F}^-$ and $^{18}\text{F}^-$. *Nature* **534**, 369–373 (2016).
6. Neumann, C. N. & Ritter, T. Facile C–F bond formation through a concerted nucleophilic aromatic substitution mediated by the PhenoFluor reagent. *Acc. Chem. Res.* **50**, 2822–2833 (2017).
7. Lucchini, V., Modena, G. & Pasquato, L. An authentic case of in-plane nucleophilic vinylic substitution: the anionotropic rearrangement of di-*tert*-butyl thirenium ions into thietium ions. *J. Am. Chem. Soc.* **115**, 4527–4531 (1993).
8. Glukhovtsev, M. N., Pross, A. & Radom, L. Is $\text{S}_{\text{N}}2$ substitution with inversion of configuration at vinylic carbon feasible? *J. Am. Chem. Soc.* **116**, 5961–5962 (1994).
9. Lucchini, V., Modena, G. & Pasquato, L. $\text{S}_{\text{N}}2$ and $\text{A}_{\text{N}}\text{E}$ mechanisms in bimolecular nucleophilic substitutions at vinyl carbon: the relevance of the LUMO symmetry of the electrophile. *J. Am. Chem. Soc.* **117**, 2297–2300 (1995).
10. Okayama, T., Takino, T., Sato, K. & Ochiai, M. In-plane vinylic $\text{S}_{\text{N}}2$ substitution and intramolecular β -elimination of β -alkylvinyl(chloro)- λ^3 -iodanes. *J. Am. Chem. Soc.* **120**, 2275–2282 (1998).
11. Bach, R. D., Baboul, A. G. & Schlegel, H. B. Inversion vs. retention of configuration for nucleophilic substitution at vinylic carbon. *J. Am. Chem. Soc.* **123**, 5787–5793 (2001).
12. Williams, A. Concerted mechanisms of acyl group transfer reactions in solution. *Acc. Chem. Res.* **22**, 387–392 (1989).
13. Curran, T. P., Farrar, C. R., Niazy, O. & Williams, A. Structure activity studies on the equilibrium reaction between phenolate ions and 2-aryloxazolin-5-one—data consistent with a concerted acyl group transfer mechanism. *J. Am. Chem. Soc.* **102**, 6828–6837 (1980).
14. Chrystiuk, E. & Williams, A. A single transition state in the transfer of methoxycarbonyl group between isoquinoline and substituted pyridines. *J. Am. Chem. Soc.* **109**, 3040–3046 (1987).
15. Ba-Saif, S. A., Luthra, A. K. & Williams, A. Concertedness in acyl group transfer: a single transition state in acetyl transfer between phenolate ion nucleophiles. *J. Am. Chem. Soc.* **109**, 6362–6368 (1987).
16. Ba-Saif, S. A., Luthra, A. K. & Williams, A. Concerted acetyl group transfer between substituted phenolate ion nucleophiles: variation of transition state structure as a function of substituent. *J. Am. Chem. Soc.* **111**, 2647–2652 (1989).
17. Han, C. & Braumann, J. I. Gas phase nucleophilic displacement reactions of negative ions with carbonyl compounds. *J. Am. Chem. Soc.* **101**, 3715–3724 (1979).
18. Kim, J. K. & Caserio, M. C. Acyl-transfer reactions in the gas phase: the question of tetrahedral intermediates. *J. Am. Chem. Soc.* **103**, 2124–2127 (1981).
19. Guthrie, J. P. Concerted mechanism for alcoholysis of esters: an examination of the requirements. *J. Am. Chem. Soc.* **113**, 3941–3949 (1991).
20. Guthrie, J. P. & Pike, D. C. Hydration of acylimidazoles: tetrahedral intermediates in acylimidazole hydrolysis and nucleophilic attack by imidazoles on esters: the question of concerted mechanisms for acyl transfer. *Can. J. Chem.* **65**, 1951–1969 (1987).
21. Hengge, A. C. & Hess, R. A. concerted or stepwise mechanisms for acyl transfer reactions of *p*-nitrophenyl acetate? transition state structures from isotope effects. *J. Am. Chem. Soc.* **116**, 11256–11263 (1994).
22. Blake, J. F. & Jorgensen, W. L. *Ab initio* study of the displacement reactions of chloride ion with formyl and acetyl chloride. *J. Am. Chem. Soc.* **109**, 3856–3861 (1987).
23. Fox, J. M., Dmitrenko, O., Liao, L. & Bach, R. D. Computational studies of nucleophilic substitution at carbonyl carbon: the $\text{S}_{\text{N}}2$ mechanism versus the tetrahedral intermediate in organic synthesis. *J. Org. Chem.* **69**, 7317–7328 (2004).
24. Renfrew, A. H. M., Taylor, J. A., Whitmore, J. M. J. & Williams, A. A single transition state in nucleophilic aromatic substitution: reaction of phenolate ions with 2-(4-nitrophenoxy)-4,6-dimethoxy-1,3,5-triazine in aqueous solution. *J. Chem. Soc. Perkin Trans. 2*, 1703–1704 (1993).
25. Xu, S. et al. The DMAP-catalyzed acetylation of alcohols—a mechanistic study. *Chem. Eur. J.* **11**, 4751–4757 (2005).
26. Skoog, M. T. & Jencks, W. P. Reactions of pyridines and primary amines with *N*-phosphorylated pyridines. *J. Am. Chem. Soc.* **106**, 7597–7606 (1984).
27. Bourne, N., Chrystiuk, E., Davis, A. M. & Williams, A. A single transition state in the reaction of aryl diphenylphosphinate esters with phenolate ions in aqueous solution. *J. Am. Chem. Soc.* **110**, 1890–1895 (1988).
28. Bourne, N. & Williams, A. Evidence for a single transition state in the transfer of the phosphoryl group to nitrogen nucleophiles from pyridino-*N*-phosphonates. *J. Am. Chem. Soc.* **106**, 7591–7596 (1984).
29. Buchwald, S. L., Friedman, J. M. & Knowles, J. R. Stereochemistry of nucleophilic displacement on two phosphoric monoesters and a phosphoguanidine: the role of metaphosphate. *J. Am. Chem. Soc.* **106**, 4911–4916 (1984).
30. Andersen, K. K., Caret, R. I. & Karup-Nielsen, I. Nucleophilic substitution at tricoordinate sulfur(IV): stereochemistry of dialkylarylsulfonium salt formation from alkyl aryl sulfoxides. *J. Am. Chem. Soc.* **96**, 8026–8032 (1974).
31. Bourne, N., Hopkins, A. & Williams, A. Single transition state for sulfonyl group transfer between pyridine nucleophiles. *J. Am. Chem. Soc.* **107**, 4327–4331 (1985).
32. D’Rozario, P., Smyth, R. L. & Williams, A. Evidence for a single transition state in the intramolecular transfer of a sulfonyl group between oxyanion donor and acceptors. *J. Am. Chem. Soc.* **106**, 5027–5028 (1984).
33. Deacon, T., Farrar, C. R., Sikkel, B. J. & Williams, A. Reactions of nucleophiles with strained cyclic sulfonate esters: Bronsted relationships for rate and equilibrium constants for variation of phenolate anion nucleophile and leaving group. *J. Am. Chem. Soc.* **100**, 2525–2534 (1978).
34. Koh, H.-J. & Um, I.-H. Kinetic study on quinuclidinolysis of *O*-phenyl *O*-Y-substituted-phenyl thionocarbonates: effects of changing nonleaving group from thionobenzoyl to phenyloxylthionocarbonyl on reactivity and transition-state structure. *Bull. Korean Chem. Soc.* **38**, 1091–1096 (2017).
35. Renfrew, A. H. M., Rettura, D., Taylor, J. A., Whitmore, J. M. J. & Williams, A. Stepwise versus concerted mechanisms at trigonal carbon: transfer of the 1,3,5-triazinyl group between aryl oxide ions in aqueous solution. *J. Am. Chem. Soc.* **117**, 5484–5491 (1995).
36. Cullum, N. R. et al. Effective charge on the nucleophile and leaving group during the stepwise transfer of the triazinyl group between pyridines in aqueous solution. *J. Am. Chem. Soc.* **117**, 9200–9205 (1995).
37. Renfrew, A. H. M., Taylor, J. A., Whitmore, J. M. J. & Williams, A. Timing of bonding changes in fundamental reactions in solution: pyridinolysis of a triazinylpyridinium salt. *J. Chem. Soc. Perkin Trans. 2* **0**, 2383–2384 (1994).
38. Kikushima, K., Grellier, M., Ohashi, M. & Ogoshi, S. Transition-metal-free hydrodefluorination of polyfluoroarenes by a concerted nucleophilic aromatic substitution with a hydrosilicate. *Angew. Chem. Int. Ed.* **56**, 16191–16196 (2017).
39. Ong, D. Y., Tejo, C., Xu, K., Hirao, H. & Chiba, S. Hydrodehalogenation of haloarenes by a sodium hydride–iodide composite. *Angew. Chem. Int. Ed.* **56**, 1840–1844 (2017).
40. Sun, H. & DiMaggio, S. Room-temperature nucleophilic aromatic fluorination: experimental and theoretical studies. *Angew. Chem. Int. Ed.* **45**, 2720–2725 (2006).
41. Zheng, Y.-J. & Bruce, T. C. On the dehalogenation mechanism of 4-chlorobenzoyl CoA by 4-chlorobenzoyl CoA dehalogenase: insights from study on the nonenzymatic reaction. *J. Am. Chem. Soc.* **119**, 3868–3877 (1997).
42. Baker, J. & Muir, M. The Meisenheimer model for predicting the principal site for nucleophilic substitution in aromatic perfluorocarbons—generalization to include ring-nitrogen atoms and non-fluorine ring substituents. *Can. J. Chem.* **88**, 588–597 (2010).
43. Goryunov, L. et al. Di- and tri-fluorobenzenes in reactions with Me_2EM ($\text{E} = \text{P}, \text{N}$; $\text{M} = \text{SiMe}_3, \text{SnMe}_3, \text{Li}$) reagents: evidence for a concerted mechanism of aromatic nucleophilic substitution. *Eur. J. Org. Chem.* **2010**, 1111–1123 (2010).
44. Cairns, A. G., Senn, H. M., Murphy, M. P. & Hartley, R. C. Expanding the palette of phenanthridinium cations. *Chem. Eur. J.* **20**, 3742–3751 (2014).
45. Glukhovtsev, M. N., Bach, R. D. & Laiter, S. Single-step and multistep mechanisms of aromatic nucleophilic substitution of halobenzenes and halonitrobenzenes with halide anions: *ab initio* computational study. *J. Org. Chem.* **62**, 4036–4046 (1997).
46. Girolardo, T., Xavier, L. A. & Riveros, J. M. An unusually fast nucleophilic aromatic displacement reaction: the gas-phase reaction of fluoride ions with nitrobenzene. *Angew. Chem. Int. Ed.* **43**, 3588–3590 (2004).
47. Fernández, I., Frenking, G. & Uggerud, E. Rate-determining factors in nucleophilic aromatic substitution reactions. *J. Org. Chem.* **75**, 2971–2980 (2010).
48. Liljenberg, M. et al. Predicting regioselectivity in nucleophilic aromatic substitution. *J. Org. Chem.* **77**, 3262–3269 (2012).
49. Liljenberg, M., Brinck, T., Rein, T. & Svensson, M. Utilizing the σ -complex stability for quantifying reactivity in nucleophilic substitutions of aromatic fluorides. *Beil. J. Org. Chem.* **9**, 791–799 (2013).
50. Clayden, J., Greeves, N., Warren, S. in *Organic Chemistry*, 2nd edn, 518 (Oxford Univ. Press, Oxford, 2012).
51. Persson, J., Axelsson, S. & Matsson, O. Solvent dependent leaving group fluorine kinetic isotope effect in a nucleophilic aromatic substitution reaction. *J. Am. Chem. Soc.* **118**, 20–23 (1996).
52. Singleton, D. A. & Thomas, A. A. High-precision simultaneous determination of multiple small kinetic isotope effects at natural abundance. *J. Am. Chem. Soc.* **117**, 9357–9358 (1995).
53. Claridge, T. D. *High-Resolution NMR Techniques in Organic Chemistry* (Elsevier, New York, NY, 2016).
54. Chan, J., Tang, A. & Bennett, A. J. A stepwise solvent-promoted $\text{S}_{\text{N}}\text{i}$ reaction of α -D-glucopyranosyl fluoride: mechanistic implications for retaining glycosyltransferases. *J. Am. Chem. Soc.* **134**, 1212–1220 (2012).
55. Westaway, K. C. Determining transition state structure using kinetic isotope effects. *J. Label. Compd Radiopharm.* **50**, 989–1005 (2007).

56. Matsson, O., Dybala-Defratyka, A., Rostkowski, M., Paneth, P. & Westaway, K. C. A theoretical investigation of α -carbon kinetic isotope effects and their relationship to the transition-state structure of S_N2 reactions. *J. Org. Chem.* **10**, 4022–4027 (2005).
57. Kwan, E. E., Park, Y., Besser, H. A., Anderson, T. L. & Jacobsen, E. N. Sensitive and accurate ^{13}C kinetic isotope effect measurements enabled by polarization transfer. *J. Am. Chem. Soc.* **139**, 43–46 (2017).
58. Papajak, E., Zheng, J., Xu, X., Leverentz, H. R. & Truhlar, D. G. Perspectives on basis sets beautiful: seasonal plantings of diffuse basis functions. *J. Chem. Theory Comput.* **7**, 3027–3034 (2011).
59. Riplinger, C. & Neese, F. An efficient and near linear scaling pair natural orbital based local coupled cluster method. *J. Chem. Phys.* **138**, 034106 (2013).
60. Karplus, M., Porter, R. & Sharma, R. Exchange reactions with activation energy. I. Simple barrier potential for (H, H_2) . *J. Chem. Phys.* **43**, 3259–3287 (1965).
61. Melander, L. C. & Saunders, W. H. *Reaction Rates of Isotopic Molecules* (Wiley, New York, NY, 1980).
62. Marcus, R. A. & Sutin, N. Electron transfers in chemistry and biology. *Biochim. Biophys. Acta* **811**, 265–322 (1985).
63. Shaik, S. S. & Hiberty, P. C. *A Chemist's Guide to Valence Bond Theory* (Wiley, New York, NY, 2007).
64. Silverstein, T. P. Marcus theory: thermodynamics can control the kinetics of electron transfer reactions. *J. Chem. Educ.* **89**, 1159–1167 (2012).
65. Bunnett, J. F. & Zahler, R. E. Aromatic nucleophilic substitution reactions. *Chem. Rev.* **49**, 273–412 (1951).
66. Marenich, A. V., Jerome, S. V., Cramer, C. J. & Truhlar, D. G. Charge model 5: an extension of Hirshfeld population analysis for the accurate description of molecular interactions in gaseous and condensed phases. *J. Chem. Theory Comput.* **8**, 527–541 (2012).
67. Chen, Z., Wannere, C. S., Corminboeuf, C., Puchta, R. & Schleyer, P. V. R. Nucleus-independent chemical shifts (NICS) as an aromaticity criterion. *Chem. Rev.* **105**, 3842–3888 (2005).
68. Glendening, E. D. et al. *NBO 6.0* (Theoretical Chemistry Institute, 2013); <http://nbo6.chem.wisc.edu/>
69. Jackson, C. J. & Gazzolo, F. H. *Am. Chem. J.* **23**, 376 (1900).
70. Meisenheimer, J. Ueber reactionen aromatischer nitrokorper. *Justus Liebigs Ann. Chem.* **323**, 205–246 (1902).
71. Helmus, J. J. *nmrglue* www.nmrglue.com
72. Frisch, M. J. et al. *Gaussian 09 and 16* (Gaussian Inc.).
73. Kwan, E. E. & Anderson, T. L. *PyQuiver* www.github.com/ekwan/pyquiver
74. Zheng, J. et al. *GAUSSRATE 2016* (University of Minnesota).
75. Zheng, J. et al. *POLYRATE 2016* (University of Minnesota).
76. Neese, F. The ORCA program system. *Wiley Inter. Rev. Comp. Mol. Sci.* **2**, 73–78 (2012).
77. Takano, Y. & Houk, K. N. Benchmarking the conductor-like polarizable continuum model (CPCM) for aqueous solvation free energies of neutral and ionic organic molecules. *J. Chem. Theory Comput.* **1**, 70–77 (2005).
78. Kongsted, J. & Mennucci, B. How to model solvent effects on molecular properties using quantum chemistry? Insights from polarizable discrete or continuum solvation models. *J. Phys. Chem. A* **111**, 9890–9900 (2007).
79. Cappelli, C., Monti, S., Scalmani, G. & Barone, V. On the calculation of vibrational frequencies for molecules in solution beyond the harmonic approximation. *J. Chem. Theory Comput.* **6**, 1660–1669 (2010).
80. Zimmerman, P. Reliable transition state searches integrated with the growing string method. *J. Chem. Theory Comput.* **9**, 3043–3050 (2013).

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Author contributions

E.E.K., Y.Z. and H.A.B. developed the isotope effect methodology. Y.Z. synthesized the materials. E.E.K. and H.A.B. carried out the calculations. E.E.K. and E.N.J. wrote the manuscript. E.N.J. guided the research.

Competing interests

The authors declare no competing interests.

Additional information

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